The Ranking Transfers, the Threshold Does Not: Early At-Risk Prediction in Unseen Courses

Doudou Coly, El Hadji Bassirou Touré and El Hadji Mamadou Nguer

***Abstract*—Early-warning systems in online higher education are usually evaluated on splits that keep the same courses in training and test sets. This methodological study asks which parts of an early at-risk model survive the move to a course it has never seen. Using the Open University Learning Analytics Dataset (32,593 enrolments, 7 modules of one university), we predict withdrawal or failure from the first quarter of each course and hold out one module at a time, with no student on both sides of the split. An audit first shows that three common evaluation choices more than double the apparent transfer gap. We then separate ranking from alert volume and calibration. Ranking largely transfers: the AUC lost on an unseen module is 0.014 on average. Calibration does not: in one module the share of students flagged rises from 44% to 95% for an observed risk of 37%, and the drift persists at every threshold from 0.3 to 0.7. Signed SHAP attributions show why: the transferred model uses the same features as local models but reads engagement and early scores on absolute scales that depend on course design. A prevalence correction does not help, but normalizing these features within the cohort, without labels, brings the alert volume back to the observed risk as well as recalibration on labelled students does, with higher precision; it needs cohorts of about 50 students and is best done with percentile ranks. The drift also creates a gender disparity through course composition, which within-course audits reveal. Transfer across institutions remains to be tested.**

***Index Terms*—Learning analytics, at-risk prediction, dropout, early warning systems, transferability, dataset shift, calibration, SHAP, algorithmic fairness, OULAD.**

I. Introduction

DROPOUT and failure remain major obstacles in online higher education. Learning management systems (LMS) record student activity from the first day of a course, which has made data-driven early-warning systems an attractive way to target support while there is still time to act. The Open University Learning Analytics Dataset (OULAD) [1] has become the reference benchmark for this task: it is open, large, and combines demographics, registration, virtual learning environment (VLE) clickstreams and assessments for 22 module presentations, and a steady stream of predictive studies builds on it [2].

High accuracy on OULAD is no longer informative in itself. Two limitations recur. First, many studies compute predictors over the whole presentation, which makes them retrospective [3]. Second, models are almost always evaluated on splits that keep the same courses on both sides; a 2026 study predicting outcomes from the first 25% of each presentation names its single-institution scope as a limit [4], and a calibrated early-warning study on OULAD holds out presentations, so other presentations of a test module can appear in training [5]. Outside OULAD, there is strong evidence that predictive models are course-specific [6]–[8], yet the mechanism of the failure is rarely examined.

These questions matter to any institution that would use a model in courses it was not trained on, for instance a newly opened programme without labelled history. We treat them as a methodological question on OULAD, where each of seven modules can play the role of an unseen course: what part of an early at-risk model's behaviour survives the move to a course it has never seen, why, and what can be done before labels exist. Our contributions are, in order of importance:

1) *An evaluation protocol and an audit.* We define a strict evaluation—prediction at a cut-off, students still enrolled, no student on both sides of a module hold-out, the same aggregation for all protocols—and show that three common choices more than double the apparent transfer gap (Table I).
2) *A diagnosis.* We separate module difficulty, the transfer cost on ranking, and the drift of alert volume and calibration. Ranking largely transfers; alert volume and calibration do not, whatever the threshold, the learner or the weighting scheme; and the modules that look hardest are difficult rather than untransferable (Sections IV-C and IV-D).
3) *A mechanism.* Signed SHAP attributions show that the transferred model relies on the same features as local ones but reads them on absolute scales whose meaning depends on course design. Global importance cannot reveal this (Section IV-F).
4) *A fairness consequence.* Calibration drift across courses converts into a gender disparity through course composition, which pooled audits read as discrimination; disparities linked to deprivation, in contrast, are present within courses (Section IV-J).
5) *A remedy that follows from the diagnosis.* Since the problem is one of scale, not of prevalence, normalizing engagement and early-assessment features within the cohort corrects it without labels; we compare it on the same students with recalibration on labelled students, with prevalence correction and with flagging a fixed share, and state the conditions it requires (Sections IV-G and IV-H).

We distinguish three levels of claims. The experiments establish

D. Coly and E. H. M. Nguer are with the Université Numérique Cheikh Hamidou Kane (UN-CHK), Dakar, Senegal (e-mail: [**TO DO:** adresses de D. Coly et E. H. M. Nguer]).

E. H. B. Touré is with the Université Cheikh Anta Diop de Dakar (UCAD), Dakar, Senegal (e-mail: elhadjibassirou.toure@ucad.edu.sn).

Corresponding author: [**TO DO:** auteur correspondant et adresse].

these results for the students of seven modules of one university; they support the interpretation that cohort normalization reduces the sensitivity of a model to course design; whether this carries over to other institutions is a hypothesis that the present data cannot test.

II. Related Work

A. Early prediction on OULAD

OULAD was released to support the at-risk prediction work of the Open University [1]. Studies have since varied the observation window, for instance predicting at several percentages of the course length [9], or from the first 25% of each presentation with deep architectures [4]. Hlosta et al. [10] address a related cold-start problem by training on the running presentation itself rather than on legacy data. The closest work is that of Wu et al. [5], who use a 28-day window, calibrate a gradient-boosting model, audit subgroup fairness and study alert capacity. Our study shares the attention to calibration, alert volume and fairness, but differs on three points: we hold out entire modules rather than presentations, so that the test course is truly unseen; we explain the calibration failure with signed attributions; and we evaluate a label-free correction based on the target cohort.

B. Portability of predictive models across courses

The Open Academic Analytics Initiative showed that a model trained at one institution could be ported to others with moderate loss [11]. Course-level analyses are less optimistic: the predictive value of LMS indicators depends on instructional conditions [6], models vary substantially across courses of one institution [7], and direct transfer between Moodle courses works only under some conditions [8]. Others built course-agnostic early predictors from massive LMS logs [12] or studied transfer learning across MOOC offerings [13]. These works measure whether accuracy transfers; we decompose *what* fails to transfer and why.

C. Dataset shift, calibration and normalization

A change of context is a case of dataset shift, in which covariate, prior or concept changes can affect a classifier [14]. In clinical prediction, discrimination often survives a change of setting while calibration does not, and updating the intercept or slope on local data restores much of it [15], [16]; Platt scaling [17] is the corresponding tool in machine learning. When only the prevalence changes, it can be estimated without labels by expectation–maximization [18] or from the confusion matrix of a black-box predictor [19]. When the input distribution changes, domain adaptation methods re-estimate normalization statistics on the target domain [20] or adapt a model at test time from unlabelled target data [21]. Normalizing features within a cohort is a simple, model-agnostic instance of this last idea, and we do not claim it as a new technique; we use it as a test of the mechanism we identify, and we state when it works.

D. Fairness of predictive student models

Fairness audits of student models compare error rates and alert rates across demographic groups [22], [23]. On OULAD, recent studies document accuracy–fairness trade-offs [24], regional disparities in recall [3] and policy-dependent alert-rate gaps [5]. When base rates differ between groups, error-rate parity and calibration cannot all hold [25], [26]; equal opportunity [27] focuses on true-positive rates. We add a structural observation: aggregating over contexts can create group disparities that exist in no context, an instance of Simpson's paradox [28].

III. Data and Methods

A. Dataset and target

OULAD [1] (CC-BY 4.0) contains 32,593 enrolments in 22 presentations of 7 modules (AAA–GGG) at the Open University, UK. Each enrolment ends in Distinction, Pass, Fail or Withdrawn. Our target is the composite unfavourable outcome $y = 1$ for Withdrawn or Fail (52.8% of enrolments), since both call for the same early support; the supplementary material repeats the analyses with Withdrawn only.

B. Early window, population and features

For a presentation p of length L_p days, the cut-off day is $d_c = \mathrm{round}(c\,L_p)$ with $c \in \{0.10, 0.20, 0.25, 0.33, 0.50\}$; $c = 0.25$ is the reference. The population at d_c consists of the students registered on or before d_c and not unregistered by d_c: students who have already left cannot be helped by an early warning, and keeping them makes the task artificially easy (Section IV-A).

For each enrolment we compute eighteen predictors from data dated on or before d_c: *demographics* (gender, region, highest education, IMD band, age band, previous attempts, studied credits, disability); *registration* (registration date relative to the course start, and days registered in advance, which equals its opposite for early registrations and is zero for late ones); *VLE engagement* (total clicks, active days, distinct resources, first and last active day, clicks per active day); *early assessments* (number of assessments, and mean score of those, whose deadline *and* submission both fall on or before d_c). Missing engagement is coded as zero activity, missing first and last active days by out-of-range sentinels, and a missing early score as -1.

Leakage control: Beyond excluding the final result and the unregistration date from the predictors [29], we test the whole pipeline by invariance: removing every raw record dated after d_c (clicks, submissions, deadlines, unregistrations) must leave the feature table unchanged. The test detects, for instance, a filter on assessment deadlines alone, which lets late submissions through (4.0% of the retained submissions at $c = 0.25$).

C. Absolute and cohort-normalized features

The *absolute* feature set uses the values above, as is usual. The *relative* set replaces the eight engagement and

early-assessment features by their percentile rank within the presentation,

$$\tilde{x}_{ij} = \frac{1}{n_p}\Big(L_{ij} + \frac{E_{ij}+1}{2}\Big), \tag{1}$$

where P_p is the population of presentation p at d_c, $n_p = |P_p|$, L_{ij} counts the students of P_p whose value of feature j is lower than x_{ij}, and E_{ij} those, i included, whose value equals it. Ties take their average rank; for instance, all students without an early score share the lowest ranks, and the rank they receive grows with their share in the cohort, which is part of the correction. Demographic and registration features are unchanged.

The transformation uses no labels, but it is *transductive*: a student's features depend on the cohort observed at the same date, so a prediction requires that cohort, ranks change as students enrol or leave, and a cohort that is collectively disengaged keeps the same ranks. To test whether ranks specifically are needed, we compare them with a z-score after winsorizing at the 1st and 99th percentiles of the cohort, and with a robust standardization by the median and the interquartile range. To assess the cohort size ranks require, we recompute them in the target module within random sub-cohorts of $n \in \{10, 20, 50, 100, 200\}$ students (20 random partitions), the trained model being unchanged.

D. Models and protocols

Our main model is a random forest [30] of 300 trees with balanced class weights (scikit-learn [31], one-hot encoding of categorical variables), which flags a student when $\hat{p} \geq 0.5$. With balanced weights this rule compares the class-conditional likelihoods of a student's features, a balanced operating point that does not depend on the prevalence of the training data. Because this choice could favour or mask a calibration problem, we also report a logistic regression with balanced weights and an unweighted random forest calibrated by isotonic regression (internal three-fold cross-validation).

Protocol (A) is a five-fold split stratified on y and grouped by student; its out-of-fold predictions cover every enrolment. Protocol (B) trains on six modules and predicts the seventh. Since 6.8% of students are enrolled in several modules (up to 48.7% of the students of CCC), (B) is *strict*: students enrolled in the held-out module are removed from the training modules, so that, as in (A), no student is on both sides. The training sets of the two protocols have similar sizes (that of (B) is 0.93 to 1.22 times that of (A)), and we check that the transfer cost is unchanged when both are trained on exactly as many students (Section IV-I).

Metrics are computed on the students of each module and then averaged over modules, for both protocols: an AUC pooled over modules in (A) would reward the separation of modules with different base rates, which (B) cannot do. For each module m we define on the same students

$$\text{difficulty:} \quad D_m = \mathrm{AUC}_A(m), \tag{2}$$

$$\text{ranking cost:} \quad C_m = \mathrm{AUC}_A(m) - \mathrm{AUC}_B(m), \tag{3}$$

$$\text{flag-rate drift:} \quad \Delta_m^{\mathrm{same}} = \bar{f}_B(m) - \bar{f}_A(m), \tag{4}$$

where $\bar{f}(m)$ is the share of the module's students flagged and both terms use the same model and feature set. To compare different feature sets or remedies with each other, we also use a common reference, $\Delta_m^{\mathrm{ref}} = \bar{f}(m) - \bar{f}_A^{\mathrm{abs}}(m)$, the difference with the absolute-feature model that has seen the module. Since that reference itself over-flags some modules, our main criterion for alert volume is the distance $|\bar{f}(m) - \bar{y}(m)|$ between the share flagged and the observed at-risk rate, which requires no reference model.

E. Calibration and threshold

We report per module, for (A) and (B), the Brier score, the mean predicted risk, the calibration-in-the-large (intercept of a logistic regression of y with $\mathrm{logit}(\hat{p})$ as offset), the calibration slope, and reliability diagrams. With balanced weights the probabilities are not calibrated even in (A): we therefore establish a *degradation* of calibration under transfer, the difference (B) − (A) on the same students, not that the local model is calibrated. We repeat the drift at thresholds 0.3 to 0.7.

F. Remedies

We compare several ways of flagging the students of a held-out module: the raw transferred model; *Platt recalibration* [17] of that model on labelled students of the target module, and the cheaper *intercept-only update* of clinical prediction [15], both fitted with balanced weights; *top-q*, which flags in each cohort the share q of students with the highest scores, q being the at-risk rate of the training modules or, without labels, the prevalence estimated in each cohort by the EM procedure of Saerens et al. [18]; and the *relative* model. All rules are evaluated on the same students: either $k = 100$ students drawn at random serve as labelled sample (50 draws) and all rules are evaluated on the others, or the first presentation of each module serves as a labelled pilot cohort and all rules are evaluated on later presentations. A top-q rule using the true at-risk rate of the module gives an unreachable bound.

G. Uncertainty

Confidence intervals (95%) come from a cluster bootstrap over students with 1,000 resamples [32]; resampling students respects the dependence between their enrolments in several modules, and the same resamples are used for all configurations, which gives paired intervals. These intervals cover the sampling of students, not of modules: at the module level we report how many modules improve and a paired Wilcoxon test ($n = 7$), and we rerun both protocols with 10 seeds.

H. Interpretability

We compute SHAP values [33] with TreeSHAP [34] in its path-dependent form, in which the expected value and the conditional expectations are taken over the training data that reached each node of the trees; attributions of a transferred model are thus relative to its training modules. We explain the transferred LOMO model and a *local* model trained on the module alone (two-fold cross-fitting grouped by student), sum

TABLE I
METHODOLOGICAL AUDIT AT THE 25% CUT-OFF (RANDOM FOREST). EACH ROW ADDS ONE CORRECTION TO THE PREVIOUS ONE. AUC OF PROTOCOL (A), RANDOM SPLIT GROUPED BY STUDENT, AND (B), LEAVE-ONE-MODULE-OUT (LOMO), MEAN OVER HELD-OUT MODULES.

Evaluation setting	AUC (A)	AUC (B)	Gap
1) Reference protocol (all enrolments, pooled AUC in (A))	0.882	0.848	0.033
2) + early scores filtered on submission date	0.880	0.847	0.033
3) + students still enrolled at the cut-off only	0.815	0.787	0.028
4) + same aggregation in (A) and (B)	0.800	0.787	0.014
5) + context-relative features	0.816	0.808	0.008

the attributions of one-hot columns back to their variable, and explain 400 students per module. Exact TreeSHAP on unpruned forests is slow; since the SHAP value of a forest is the average of those of its trees, we estimate it without bias from 100 trees drawn at random, which reproduces the exact importance ranking with Spearman $\rho = 0.985$. The *agreement* between transferred and local models is the Spearman correlation of their global importances (mean $|\phi|$); the *drift decomposition* averages signed attributions of the transferred model over the students of m and over feature families G,

$$\bar{\phi}_{m,G} = \frac{1}{|S_m|} \sum_{i \in S_m} \sum_{j \in G} \phi_{ij}, \tag{5}$$

whose sum over families equals the mean difference between the predicted risk of m's students and the average prediction on the training modules.

I. Fairness

We compare true-positive rate (TPR), false-positive rate (FPR), precision (PPV) and flag rate between women and men, the most and least deprived IMD quintiles (Q1: 0–20%; Q5: 80–100%), and students with and without a declared disability, over all modules and within each module, with group sizes and the cluster bootstrap. To formalize composition effects, we regress the flag of students not at risk on gender with and without module fixed effects (linear probability model). These are descriptive associations; they do not identify the causes of a disparity.

J. Implementation and reproducibility

All analyses run in Python 3.12 with scikit-learn 1.9.1, pandas 3.0.6, shap 0.54.1 and PyTorch 2.14.1, with fixed seeds. A single script regenerates every figure, table and number of the article and of its supplementary material from the raw OULAD files, and an automated test suite checks the absence of temporal leakage and that no student is on both sides of either protocol.

IV. RESULTS

A. Methodological audit

The pipeline we started from, which keeps all enrolments and compares a pooled AUC in (A) with a per-module mean in (B), gives AUC 0.882 in (A) and 0.848 in (B) at 25% of the course, a transfer gap of 0.033 (Table I; gaps are computed on unrounded values). Filtering early scores on their submission date changes little. Restricting the population to students still enrolled at the cut-off has the largest effect: at 25%, 19.5% of enrolments have already ended, 62.5% of all Withdrawn outcomes among them, including 2,678 unregistrations before the course started. Their last active day makes them trivial to predict, and their share ranges from 5.4% (GGG) to 28.0% (CCC), which confounds comparisons between modules. Comparing like with like halves the gap again, from 0.028 to 0.014. Together, these choices make the apparent gap more than twice its corrected value. They also explain why our AUCs at 25% of the course, around 0.80, are lower than the values above 0.90 often reported on OULAD: those are typically obtained with features covering the whole course, or with students who have already left.

B. How early can we predict?

Fig. 1 shows performance against the observed window. With relative features, AUC rises from 0.749 at 10% of the course to 0.878 at 50% in (A), and from 0.739 to 0.871 in (B), with a gap between protocols of 0.007 to 0.010 at every cut-off (0.012–0.017 with absolute features). With absolute features, the transferred model flags too many students: at 25%, its recall exceeds that of (A) (0.719 versus 0.649) while its precision falls (0.618 versus 0.669). Reporting recall alone, a common practice, would make the transferred model look better than the local one.

C. The ranking transfers, the alert volume does not

Table III and Fig. 2 separate the three quantities per module at 25%. The ranking cost is small: 0.014 [0.010, 0.017] on average with absolute features, and below 0.015 in 6 modules out of seven. Modules AAA and GGG, which a naive reading of LOMO results would call "poorly transferable", are in fact *difficult*: their AUC is already low when the model has seen them (0.733 and 0.739) and drops by only 0.002 and 0.014 when it has not. BBB is the exception, with a cost of 0.043, which persists within its presentations (0.033) and which relative features reduce to 0.011: its ranking, too, depends on scale.

The alert volume is another matter. In GGG, the transferred model flags 95% of students, against 44% for the model that has seen the module and an observed risk of 37%: recall climbs from 0.66 to 0.99 and precision falls from 0.55 to 0.38. BBB drifts too (+0.19), CCC and DDD slightly downwards; the mean absolute drift $|\Delta^{\text{same}}|$ is 0.135 [0.130, 0.141]. An institution that deployed such a model in a GGG-like course would contact almost every student.

D. A calibration problem, not only a threshold

The drift is not an artefact of the threshold. With absolute features, the mean drift stays between 0.084 and 0.138 for every threshold from 0.3 to 0.7, and the threshold-free shift in mean predicted risk is 0.075 on average and +0.30 for GGG. Calibration degrades accordingly (Table IV): moving to an

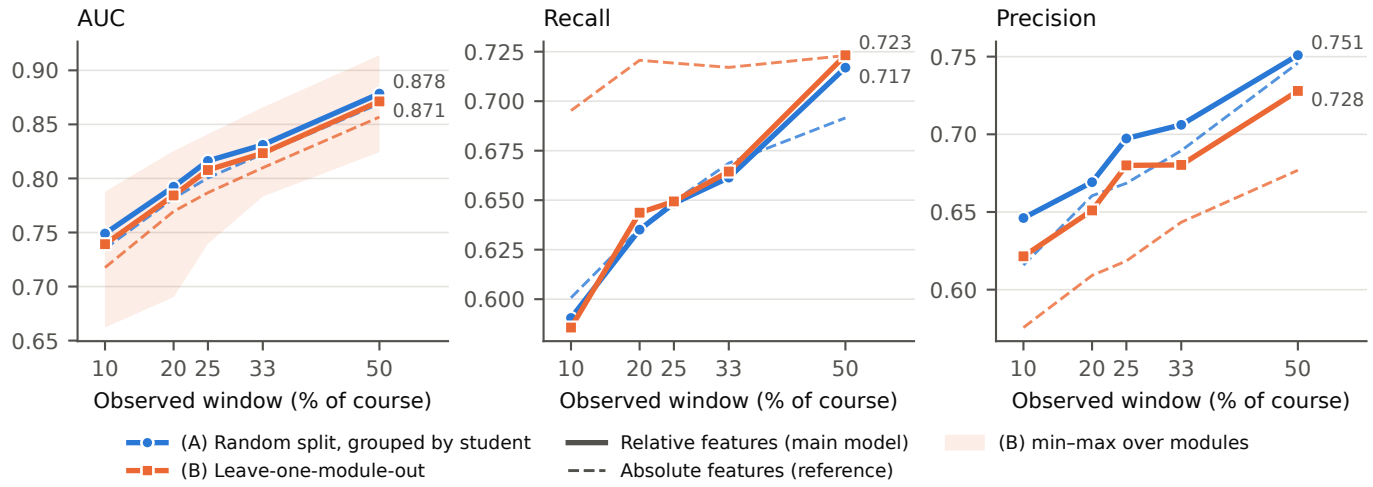


Fig. 1. Performance against the observed share of the presentation (students still enrolled at the cut-off). Solid lines: relative features; dashed lines: absolute features. Both protocols are averaged over modules in the same way; the band shows the range over held-out modules in (B) with relative features.

TABLE II

MODULE CHARACTERISTICS AT THE 25% CUT-OFF, STUDENTS STILL ENROLLED AT THE CUT-OFF. AT-RISK: WITHDRAWN OR FAIL. FIRST DEADLINE: FIRST NON-EXAM ASSESSMENT, AS A FRACTION OF THE PRESENTATION LENGTH. EXAM WEIGHT: SHARE OF THE EXAM IN THE TOTAL ASSESSMENT WEIGHT. EARLY SCORE: SHARE OF STUDENTS WITH AT LEAST ONE ASSESSMENT SUBMITTED BEFORE THE CUT-OFF. VALUES ARE AVERAGED OVER THE PRESENTATIONS OF EACH MODULE.

Module	n	At-risk (%)	Withdrawn among at-risk (%)	First deadline	Exam weight (%)	Early score (%)	Median clicks	Aged 35+ (%)
AAA	698	23.9	45.5	0.07	50	98.1	588	52.6
BBB	6,202	39.5	28.0	0.07	50	93.4	224	34.9
CCC	3,191	47.4	48.4	0.07	67	93.1	402	30.3
DDD	4,984	47.7	40.7	0.09	50	93.4	382	26.2
EEE	2,470	33.2	31.6	0.13	50	90.9	586	24.9
FFF	6,289	42.0	35.4	0.08	50	94.5	866	26.2
GGG	2,398	36.9	17.6	0.24	100	53.3	81	39.0

TABLE III

RANKING VERSUS THRESHOLD TRANSFER PER MODULE (25% CUT-OFF, STUDENTS STILL ENROLLED). AUC AND FLAG RATE ($p \geq 0.5$) ARE COMPUTED ON THE STUDENTS OF EACH MODULE; (A) USES OUT-OF-FOLD PREDICTIONS OF THE RANDOM SPLIT, (B) THE MODEL TRAINED ON THE SIX OTHER MODULES.

	Ranking (absolute features)			Flagged students, % (absolute)			Relative features, (B)	
Module	AUC (A)	AUC (B)	Cost	Observed risk	(A)	(B)	AUC	Flagged (%)
AAA	0.733	0.731	+0.002	24	35	45	0.792	37
BBB	0.796	0.753	+0.043	39	37	56	0.801	40
CCC	0.830	0.817	+0.012	47	44	41	0.811	46
DDD	0.837	0.827	+0.010	48	44	39	0.841	39
EEE	0.841	0.838	+0.003	33	26	31	0.837	29
FFF	0.828	0.817	+0.011	42	36	33	0.833	36
GGG	0.739	0.725	+0.014	37	44	95	0.739	30
Mean	0.800	0.787	+0.014	39	38	49	0.808	37

unseen module changes the Brier score by 0.039 on average (+0.180 for GGG), the calibration-in-the-large by 0.42 (-1.83 for GGG) and the slope of GGG by -0.321. The reliability diagrams make the failure visible (Fig. 3): over all modules, the three models are close; in GGG, the transferred absolute model predicts risks far above the observed ones, while the relative model stays with the local one. With relative features, the changes shrink to 0.006, 0.08 and 0.077 on average.

The choice of balanced weights is not the cause. An unweighted forest calibrated by isotonic regression drifts as much on unseen modules ($|\Delta^{\text{same}}| = 0.150$; mean predicted risk 0.139 away from the observed rate), and so does, to a lesser extent, a logistic regression (0.061; Table VII).

E. Why GGG drifts

Table II shows how GGG differs. Its first non-exam assessment falls at about a quarter of the presentation instead of 7–13%, its assignments are formative (the exam carries the whole weight), and only half of its students have an early score at the cut-off. Its students also use the VLE much less: their median click count before the cut-off is 81 against 432 in

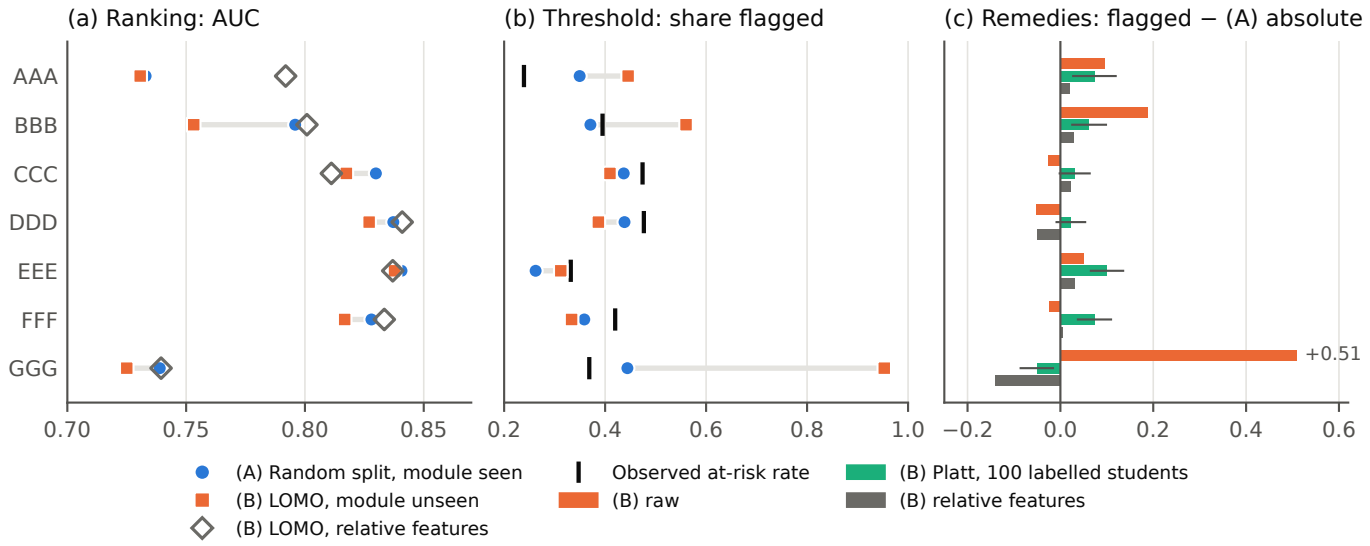


Fig. 2. Ranking versus alert volume per module at the 25% cut-off. (a) AUC when the module is seen (A) or unseen (B), and (B) with relative features. (b) Share of students flagged in (A) and (B) against the observed at-risk rate. (c) Remedies on the same students, against the common reference (Δ^{ref}): raw transferred model, Platt recalibration on 100 labelled students (mean and standard deviation over 50 draws), and relative features.

TABLE IV

CALIBRATION OF THE TRANSFERRED MODEL AT THE 25% CUT-OFF: DIFFERENCE BETWEEN PROTOCOL (B) AND PROTOCOL (A) ON THE SAME STUDENTS, AS THE MEAN ABSOLUTE DIFFERENCE OVER MODULES AND FOR MODULE GGG. BRIER SCORE; CALIBRATION-IN-THE-LARGE (INTERCEPT OF A LOGISTIC RECALIBRATION WITH $\text{logit}(\hat{p})$ AS OFFSET); MEAN PREDICTED RISK; CALIBRATION SLOPE. WITH BALANCED CLASS WEIGHTS THE PROBABILITIES ARE NOT CALIBRATED EVEN IN (A), SO THE TRANSFER EFFECT IS THE DIFFERENCE (B) − (A).

	Brier		Calibration-in-the-large		Mean predicted risk		Slope	
Features	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG	mean $\lvert\Delta\rvert$	GGG
Absolute	0.039	+0.180	0.419	−1.834	0.075	+0.298	0.170	−0.321
Relative	0.006	+0.004	0.079	−0.061	0.015	+0.011	0.077	−0.094

the other modules, the 11th percentile of the latter. Elsewhere, students with at most 81 clicks are at risk in 79% of cases; in GGG, 47%. Elsewhere, a student without an early score is at risk in 98% of cases; in GGG, where 47% of students have none yet, 51%. The drift is not limited to the period before the first deadline: it reaches +0.60 at 33% and +0.51 at 50% of the course. The model has learned absolute thresholds—"few clicks" or "no score yet" means risk—whose meaning depends on course design. Across modules and presentations, drift is larger where at-risk students fail rather than withdraw (Spearman $\rho = -0.68$ over presentations), where students are older and less qualified, and where the VLE is used less; with seven modules these associations are exploratory (Table S5 of the supplementary material).

F. Interpretability: same features, wrong scales

If transfer failed because different features matter in different modules, global importances would differ between transferred and local models. They mostly do not. Local models rank the 18 variables similarly across modules (mean pairwise Spearman 0.81), with the mean early score first everywhere. Agreement between transferred and local importances is moderate with absolute features (0.60) and higher with relative features (0.81), an improvement in 6 modules out of seven (Fig. 4a). Crucially, GGG—the module that drifts most—shows one of the *highest* agreements with absolute features (0.81). Its problem is not which features are used but how they are read.

The drift decomposition of Eq. (5) makes this visible (Fig. 4b). On GGG's students, the transferred absolute model raises predicted risk by +0.30 above its training average: +0.19 from early-score features and +0.12 from engagement, while demographics contribute −0.01. Across modules, this total follows the flag-rate drift ($\rho = 0.71$). With relative features the decomposition vanishes (−0.04 for GGG; Fig. 4c). Global importance, which discards the sign of attributions, cannot detect a calibration problem; signed attributions on the target population can.

G. Remedies compared on the same students

Table V compares the remedies on the same students. The label-free prevalence correction fails: the prevalence that the EM procedure estimates for GGG leads to flagging 100% of its students, and the resulting rule is further from the observed risk than the raw model ($|\text{flagged} - \text{risk}| = 0.224$ against 0.174). This is what the diagnosis predicts. Under a pure change of prevalence, the balanced rule of our model would need no correction; the drift comes instead from a change in how features relate to risk, which inflates the posteriors that prevalence estimation relies on.

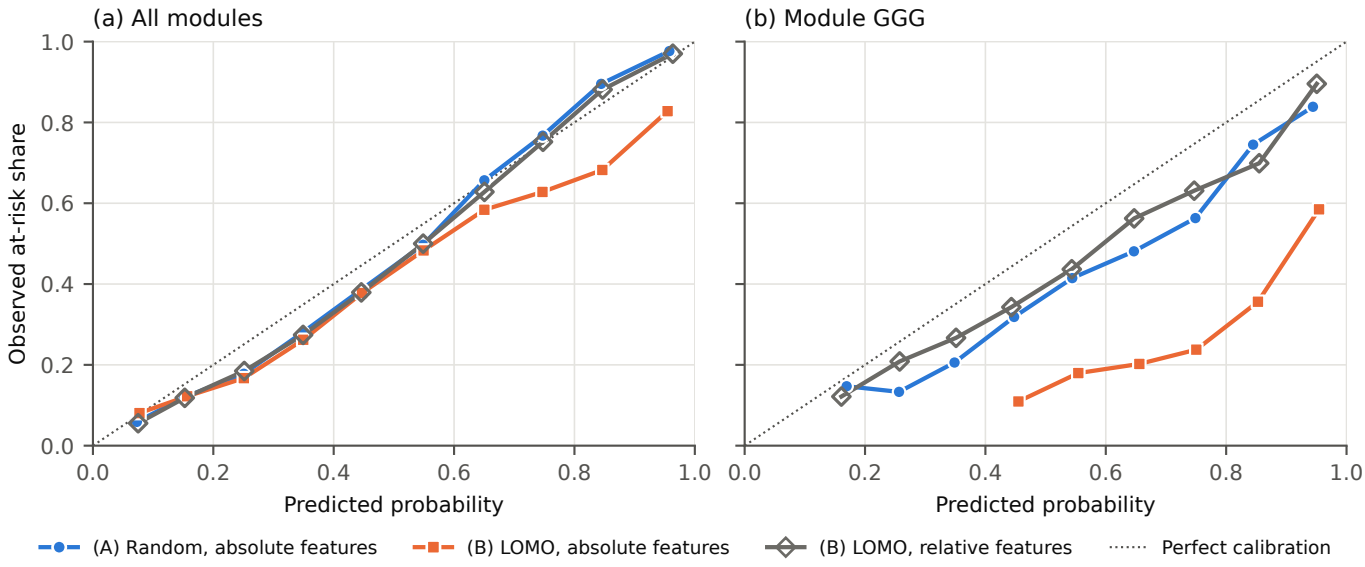


Fig. 3. Reliability diagrams at the 25% cut-off: observed at-risk share by class of predicted probability (classes of fewer than 30 students omitted). With balanced class weights, the curves need not lie on the diagonal even in (A); the transfer effect is the gap between (B) and (A).

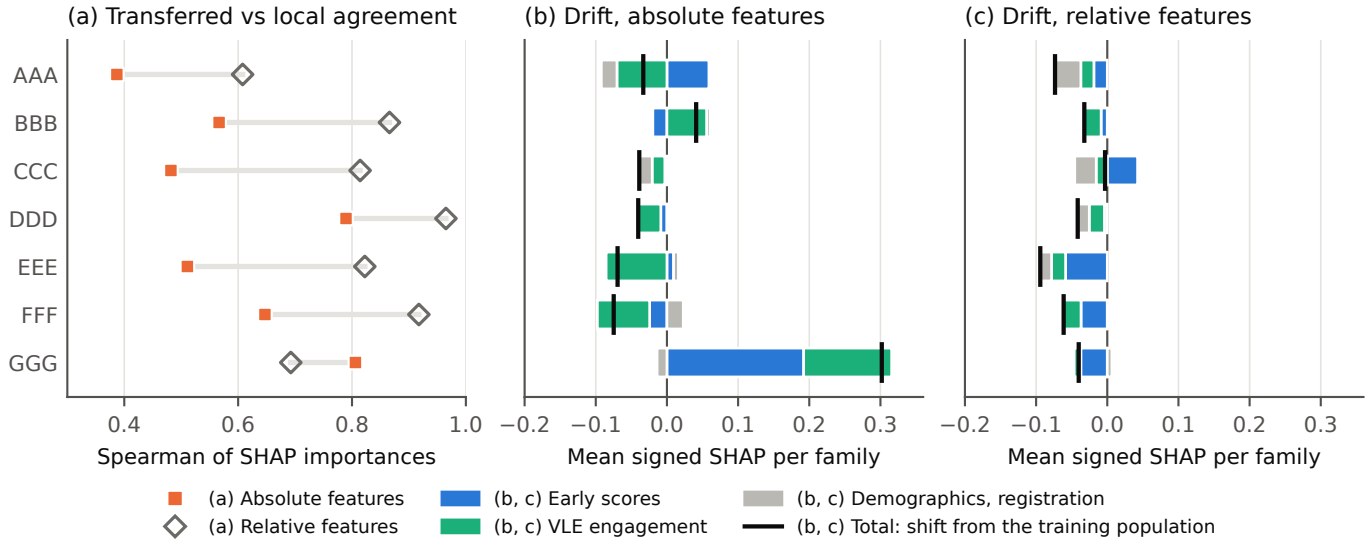


Fig. 4. Same features, wrong scales (25% cut-off). (a) Spearman correlation between the SHAP importances of the transferred model and of a model trained on the module itself. (b, c) Mean signed SHAP attribution of the transferred model on the students of each module, by feature family: the shift of their predicted risk from the training population.

With 100 labelled students, Platt recalibration and the intercept-only update both bring the alert volume close to the observed risk (0.057 and 0.058), but leave precision almost unchanged (0.624 and 0.627, against 0.619) and vary from one sample of 100 students to another: the standard deviation of the share flagged is 0.039 for Platt and 0.040 for the intercept, which is therefore not more stable. Flagging a fixed share of each cohort reaches 0.068. Without any label, the relative model is as close to the observed risk as the recalibrations (0.057, the same as the model that has seen the module, 0.059), with the highest precision (0.681) and F1 (0.659). With a labelled pilot cohort instead of random students, recalibration does worse (precision 0.615) because presentations of a module also differ, whereas the relative model keeps its advantage on the same later presentations (0.679). Against the common reference, the relative model under-flags GGG ($\Delta^{\text{ref}} = -0.14$); but that reference over-flags GGG, and the distance to the observed risk is what matters for alert volume.

H. Which normalization, and which model?

Ranks are not the only way to normalize within the cohort (Table VI). A winsorized z-score and a robust standardization reach an AUC of 0.801 and 0.800 on unseen modules, against 0.808 for ranks and 0.787 for absolute values; across three cut-offs the three normalizations differ by at most 0.008 in mean AUC. What matters is to normalize within the cohort; ranks are the best or tied option, have no parameter and are

TABLE V

REMEDIES COMPARED ON THE SAME STUDENTS (25% CUT-OFF; MEANS OVER MODULES). PLAN k: PLATT AND INTERCEPT-ONLY UPDATES USE $k = 100$ LABELLED STUDENTS OF THE TARGET MODULE DRAWN AT RANDOM (50 DRAWS; STANDARD DEVIATION OF THE SHARE FLAGGED IN PARENTHESES), AND ALL RULES ARE EVALUATED ON THE REMAINING STUDENTS. PILOT: THE FIRST PRESENTATION OF EACH MODULE IS THE LABELLED COHORT AND ALL RULES ARE EVALUATED ON LATER PRESENTATIONS. $|\Delta^{\mathrm{ref}}|$: ABSOLUTE FLAG-RATE DRIFT AGAINST THE ABSOLUTE-FEATURE MODEL WITH THE MODULE SEEN; $|\text{FLAGGED} - \text{RISK}|$: DISTANCE BETWEEN THE SHARE FLAGGED AND THE OBSERVED AT-RISK RATE, WHICH REQUIRES NO REFERENCE MODEL. TOP-q FLAGS THE q HIGHEST SCORES OF EACH COHORT, q BEING THE AT-RISK RATE OF THE TRAINING MODULES, THE RATE ESTIMATED WITHOUT LABELS BY EM [18], OR THE TRUE RATE (UNREACHABLE BOUND). PER-MODULE VALUES ARE GIVEN IN THE SUPPLEMENTARY MATERIAL.

	Target	Plan $k = 100$				Pilot cohort			
Rule	labels	$\lvert\Delta^{\mathrm{ref}}\rvert$	$\lvert\text{flagged} - \text{risk}\rvert$	Precision	F1	$\lvert\Delta^{\mathrm{ref}}\rvert$	$\lvert\text{flagged} - \text{risk}\rvert$	Precision	F1
(A) absolute, module seen	—	0.000	0.059	0.669	0.653	0.000	0.077	0.666	0.650
Raw transferred	no	0.135	0.174	0.619	0.637	0.134	0.193	0.621	0.629
Platt recalibration	k	0.059 (0.039)	0.057	0.624	0.647	0.097	0.127	0.615	0.629
Intercept-only update	k	0.051 (0.040)	0.058	0.627	0.647	0.090	0.125	0.621	0.627
Top-q (training rate)	no	0.058	0.068	0.631	0.644	0.061	0.061	0.623	0.641
Top-q (EM estimate)	no	0.187	0.224	0.623	0.585	0.205	0.273	0.640	0.559
Relative features	no	0.043	0.057	0.681	0.659	0.067	0.066	0.679	0.655
Top-q (true rate, bound)	oracle	0.060	0.003	0.648	0.647	0.088	0.013	0.640	0.642

insensitive to sentinel values. Ranks need a reasonable cohort: when they are computed in random sub-cohorts of the target module, the AUC on unseen modules is 0.778 [0.773, 0.781] with 10 students and 0.802 [0.800, 0.804] with 50, against 0.808 with the whole presentation (Table S2); from about 50 students they are as reliable as with the whole cohort.

The effect does not depend on the learner (Table VII). Ranks raise the AUC on unseen modules and reduce the drift for the logistic regression (0.775 to 0.805; 0.061 to 0.018) and for the calibrated unweighted forest (0.786 to 0.810; 0.150 to 0.034), as they do for gradient boosting and a recurrent network on weekly activity sequences (Table S6). The average precision leads to the same conclusions (0.720 to 0.755 for the forest; Table S3).

I. Uncertainty, heterogeneity and robustness

The differences between relative and absolute features are well estimated for the students of these modules: +0.021 [+0.015, +0.027] in AUC on unseen modules, -0.005 [-0.009, -0.001] in ranking cost and -0.115 [-0.120, -0.107] in mean absolute drift. At the module level they are less uniform. The drift decreases in 6 modules out of seven (Wilcoxon $p = 0.047$); the AUC improves in 5 ($p = 0.078$), so the gain in ranking is not established at the module level. It is large for AAA (+0.063) and BBB (+0.048), while CCC loses -0.008 and EEE -0.002 (means over 10 seeds, whose standard deviation is at most 0.004 per module). CCC is the module whose absolute scales already transfer best (its drift is 0.027); its loss persists within each presentation (-0.014), so it is not due to differences between presentations: ranks discard a resolution of the absolute values that was useful there, without anything to correct.

The conclusions are robust to the training size (at equal size, the transfer cost is 0.014 and 0.008), to keeping in training the students shared with the held-out module (mean AUC change at most 0.003), to the threshold, to the learner and to the target definition; the supplementary material details these checks.

J. Fairness

Table VIII and Fig. 5 report error-rate gaps between groups. With absolute features, transfer creates a gender gap that does not exist when modules are seen: the FPR gap between women and men moves from -0.007 [-0.019, +0.005] in (A) to +0.194 [+0.180, +0.208] in (B). Within modules, the transferred model's FPR is never significantly higher for women, and it is significantly lower in 5 modules out of seven (Fig. 5c). A regression with module fixed effects confirms the composition effect: the gender coefficient goes from +0.194 without fixed effects to -0.048 [-0.062, -0.033] with them. Women make up 88% of BBB and 81% of GGG, the two modules whose alert volume drifts upwards, so pooling attributes their drift to gender [28]. With relative features, the pooled gap is strongly attenuated, to +0.015 [+0.003, +0.026]; it is reduced, not eliminated.

IMD and disability gaps behave differently. They exist under every protocol and reflect different base rates (51% at risk in Q1 against 33% in Q5), and, unlike the gender gap, they are present within courses: the FPR of Q1 is significantly higher in 5 modules out of seven. Relative features reduce the pooled FPR gap by -0.039 [-0.064, -0.017] without closing it, consistent with the impossibility of equalizing error rates and calibration when base rates differ [25], [26]. Intersections show that, with absolute features, disadvantages add up (FPR 0.496 for women in Q1 against 0.169 for men in Q5), whereas with relative features the remaining gaps follow deprivation rather than gender (Table S4).

V. DISCUSSION

A. What transfers and what does not

The ordering of students by risk is largely shared across modules: early scores, recent activity and submissions predict risk everywhere. What is specific to a course is the scale on which these signals are expressed: how many clicks a normal week requires, when the first assessment is due, whether early assignments are graded. A model that learns absolute thresholds carries the scale of its training courses into the new one, and its probabilities degrade there. The parallel with clinical prediction, where discrimination travels better than calibration [15], [16], is close. The failure of prevalence correction sharpens the diagnosis: the shift is not in how many students are at risk, but in what their activity means.

TABLE VI

WHICH COHORT NORMALIZATION? THE EIGHT ENGAGEMENT AND EARLY-ASSESSMENT FEATURES ARE NORMALIZED WITHIN EACH PRESENTATION (WEIGHTED FOREST, 25% CUT-OFF, LOMO STRICT, MEANS OVER MODULES). $|\Delta^{\text{same}}|$: MEAN ABSOLUTE FLAG-RATE DRIFT (B) − (A) OF THE SAME VARIANT; |FLAGGED − RISK|: DISTANCE BETWEEN THE SHARE FLAGGED IN (B) AND THE OBSERVED AT-RISK RATE.

Normalization	AUC (A)	AUC (B)	Cost	$\lvert\Delta^{\text{same}}\rvert$	|flagged − risk|	Precision (B)	F1 (B)
Absolute values	0.800	0.787	+0.014	0.135	0.174	0.618	0.637
Percentile rank	0.816	0.808	+0.008	0.021	0.057	0.680	0.658
z-score (winsorized 1–99%)	0.814	0.801	+0.014	0.037	0.061	0.666	0.653
Robust (median, IQR)	0.809	0.800	+0.009	0.024	0.054	0.664	0.648

TABLE VII

IS THE EFFECT SPECIFIC TO THE WEIGHTED FOREST? PROTOCOL (B), 25% CUT-OFF, MEANS OVER MODULES, WITH ABSOLUTE FEATURES AND WITHIN-COHORT RANKS. WEIGHTED FOREST AND LOGISTIC REGRESSION USE BALANCED CLASS WEIGHTS AND THRESHOLD 0.5; THE UNWEIGHTED FOREST IS CALIBRATED BY ISOTONIC REGRESSION AND USES THRESHOLD 0.5 ON CALIBRATED PROBABILITIES. AP: AVERAGE PRECISION. $|\Delta^{\text{same}}|$: ABSOLUTE FLAG-RATE DRIFT AGAINST (A) WITH THE SAME MODEL AND FEATURES. $|\bar{p}-\text{RISK}|$: DISTANCE BETWEEN MEAN PREDICTED RISK AND OBSERVED AT-RISK RATE. |ΔBRIER|: ABSOLUTE CHANGE OF THE BRIER SCORE FROM (A) TO (B). CAPACITY PRECISION: PRECISION WHEN FLAGGING THE 30% HIGHEST SCORES OF EACH COHORT.

Model	Features	AUC (B)	AP (B)	$\lvert\Delta^{\text{same}}\rvert$	|flagged − risk|	$\lvert\bar{p}-\text{risk}\rvert$	|ΔBrier|	Capacity precision
Weighted forest	absolute	0.787	0.720	0.135	0.174	0.134	0.039	0.708
	rank	0.808	0.755	0.021	0.057	0.069	0.006	0.731
Logistic regression	absolute	0.775	0.716	0.061	0.148	0.123	0.022	0.694
	rank	0.805	0.751	0.018	0.079	0.094	0.004	0.720
Unweighted calibrated forest	absolute	0.786	0.720	0.150	0.191	0.139	0.038	0.708
	rank	0.810	0.756	0.034	0.112	0.052	0.006	0.732

TABLE VIII

BETWEEN-GROUP GAPS IN ERROR RATES AT THE 25% CUT-OFF (THRESHOLD 0.5), WITH 95% CONFIDENCE INTERVALS FROM A STUDENT-LEVEL CLUSTER BOOTSTRAP (1,000 RESAMPLES). TPR: RECALL; FPR: FALSE-POSITIVE RATE; PPV: PRECISION; FLAG: SHARE OF STUDENTS FLAGGED.

Contrast	Configuration	TPR gap	FPR gap	PPV gap	Flag-rate gap
Female − male	(A) absolute	−0.080 [−0.096, −0.061]	−0.007 [−0.019, +0.005]	−0.042 [−0.058, −0.025]	−0.050 [−0.062, −0.038]
	(B) absolute	+0.018 [+0.001, +0.035]	+0.194 [+0.180, +0.208]	−0.173 [−0.189, −0.155]	+0.111 [+0.099, +0.123]
	(B) relative	−0.065 [−0.082, −0.047]	+0.015 [+0.003, +0.026]	−0.060 [−0.077, −0.043]	−0.031 [−0.043, −0.020]
IMD Q1 − Q5	(A) absolute	+0.135 [+0.104, +0.165]	+0.148 [+0.127, +0.169]	+0.044 [+0.013, +0.077]	+0.224 [+0.205, +0.243]
	(B) absolute	+0.145 [+0.115, +0.175]	+0.156 [+0.132, +0.180]	+0.111 [+0.080, +0.142]	+0.219 [+0.200, +0.239]
	(B) relative	+0.125 [+0.095, +0.157]	+0.116 [+0.096, +0.137]	+0.062 [+0.032, +0.094]	+0.205 [+0.186, +0.226]
Disability yes − no	(A) absolute	+0.095 [+0.067, +0.121]	+0.102 [+0.076, +0.128]	+0.020 [−0.006, +0.047]	+0.146 [+0.124, +0.167]
	(B) absolute	+0.050 [+0.025, +0.074]	+0.111 [+0.083, +0.139]	+0.036 [+0.011, +0.064]	+0.121 [+0.099, +0.140]
	(B) relative	+0.078 [+0.051, +0.105]	+0.060 [+0.034, +0.085]	+0.045 [+0.020, +0.072]	+0.117 [+0.094, +0.138]

The three levels of claims stated in the introduction should be kept apart. The gains of cohort normalization are established for the students of seven modules, with intervals that exclude zero, but they are uneven across modules: the reduction of drift holds in six modules out of seven, the gain in ranking is not established at the module level, and CCC is slightly worse off. The interpretation that normalization reduces sensitivity to course design rests on the mechanism, the SHAP decomposition and the consistency across learners. Transfer across institutions is not tested here.

B. Implications for deployment in a new context

Four recommendations follow for an institution using a model in a course without labelled history. First, normalize engagement and early-assessment features within the cohort at the prediction date; this needs no outcome data but a cohort of at least about 50 students, so smaller groups should be pooled across sections or sessions of the same course. Second, compare the alert volume with expectations from the first weeks: a flag rate far from the expected risk is a signal to investigate, since without labels it cannot tell a miscalibrated model from a cohort that is truly more at risk. Third, once a first cohort is labelled, recalibrate, keeping in mind that recalibration on about a hundred students varies from one sample to another. Fourth, decide how alerts are used before deploying them. Since gaps linked to deprivation and disability persist within courses and cannot all be removed when base rates differ, alerts should trigger an offer of support reviewed by a person, not an automatic decision, and their distribution across groups should be monitored.

Reducing a disparity is not the same as making decisions fair. Cohort normalization removes the gender gap created by pooling courses, but it also changes who is flagged within a course, and it hides any collective disengagement of a cohort. A practical design would combine two levels: a cohort-relative model to identify students who disengage compared with their peers, and an absolute cohort-level indicator to detect a decline of the whole cohort, which would call for a pedagogical rather

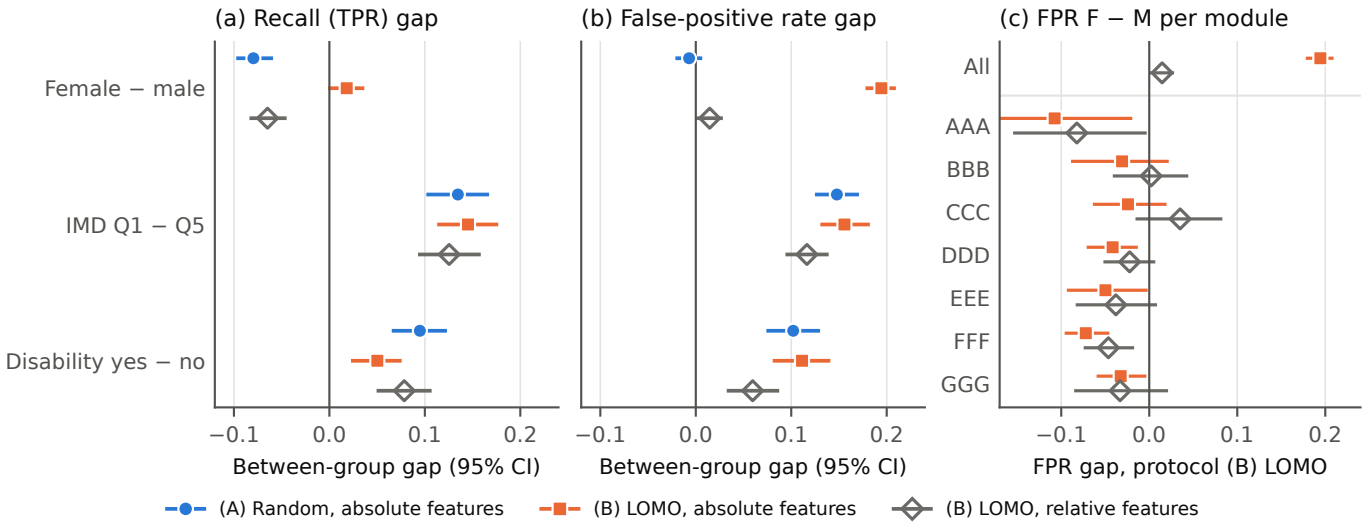


Fig. 5. Error-rate gaps between groups (25% cut-off, 95% cluster-bootstrap intervals). (a) Recall and (b) false-positive rate gaps. (c) Female–male FPR gap of the transferred model within each module and over all modules, with absolute and relative features: the pooled gap of the absolute model is a composition effect.

than an individual response.

C. Implications for evaluation practice

Five lessons generalize beyond OULAD. Early-prediction studies should restrict the population to students still enrolled at the prediction date and check leakage by invariance. Course-level hold-out should exclude students enrolled in the held-out course from training. Comparisons between protocols must use the same aggregation. Alert rates and calibration should be reported alongside discrimination, since recall alone rewards over-flagging. Fairness audits that pool contexts should be complemented by within-context audits; otherwise a calibration problem can read as discrimination and, conversely, a real within-context bias can be masked by composition.

D. Limitations

The study uses one institution and seven modules; module-level tests have little power, presentation-level associations are not independent, and our bootstrap intervals cover the sampling of students, not of courses. Our main model uses balanced weights and a fixed threshold; we checked thresholds from 0.3 to 0.7, a calibrated unweighted forest, a logistic regression, gradient boosting and a small recurrent network, but richer sequence models might extract more temporal signal. SHAP values are estimated from a subsample of trees and of students. Cohort normalization is transductive, needs cohorts of about 50 students and cannot detect a collectively disengaged cohort. IMD and disability gaps remain within courses. Finally, the transfer we study is between modules of one university; transfer across institutions and countries remains to be tested.

E. Future work

We will replay the same protocol on Moodle data from a fully online university in Senegal, where the held-out context becomes a degree programme and then a country, and test the two-level design that combines cohort-relative and absolute cohort-level indicators.

VI. Conclusion

Measured strictly—early, on students still enrolled, with no student on both sides of the split and with comparable aggregation—the cost of moving an at-risk model to an unseen course is small for ranking and large for alert volume and calibration. Calibration degrades because engagement and early scores are read on absolute scales that depend on course design, not because different features matter or because the prevalence changes. Normalizing these features within the cohort corrects the alert volume without labels, provided cohorts are large enough, and strongly attenuates a gender disparity created by pooling courses. Early-warning systems meant to travel should be evaluated on unseen contexts, report alert rates and calibration alongside discrimination, and audit fairness within contexts.

Ethics Statement

This study is a secondary analysis of OULAD, a public dataset released under a CC-BY 4.0 licence in which students are anonymized by the data provider [1]. No new data were collected, no attempt was made to re-identify students, and no intervention was carried out; under these conditions no ethical approval was required. [**TO DO:** confirmer selon les règles de l'UN-CHK et de la revue.]

Author Contributions

[**TO DO:** proposition à valider par les trois auteurs.] D. Coly: conceptualization, methodology, software, formal analysis, investigation, visualization, writing—original draft. E. H. B. Touré: supervision, methodology, writing—review and editing. E. H. M. Nguer: supervision, conceptualization, writing—review and editing.

FUNDING

[**TO DO:** financement, ou "This research received no specific grant from any funding agency".]

CONFLICTS OF INTEREST

[**TO DO:** déclaration, par exemple "The authors declare no conflict of interest".]

DATA AND CODE AVAILABILITY

OULAD is available under a CC-BY 4.0 licence from the Open University (https://analyse.kmi.open.ac.uk/open_dataset). The code that reproduces every figure, table and number of this article and of its supplementary material, with its tests, is available at https://github.com/duducolydev/analyses_oualad. [**TO DO:** DOI Zenodo de la version archivée.]

REFERENCES

[1] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open university learning analytics dataset," *Scientific Data*, vol. 4, p. 170171, 2017.

[2] L. Jin, Y. Wang, H. Song, and H. J. So, "Predictive modelling with the Open University Learning Analytics Dataset (OULAD): A systematic literature review," in *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium and Blue Sky (AIED 2024)*. Springer, 2024, pp. 477–484.

[3] A. Elsayed, Y. Wardat, F. Kamalov, and H. Sulieman, "Toward sustainable and equitable AI in education through a regional fairness audit of dropout prediction using the OULAD dataset," *Sustainability*, vol. 18, no. 18, p. 9440, 2026.

[4] C. Malkani, N. Mittal, and S. Som, "A hybrid deep Kronecker and BN-eLU-AlexNet framework for early prediction of learner outcomes in e-learning platforms," *Frontiers in Computer Science*, vol. 8, p. 1889642, 2026.

[5] S. Wu, J. Duan, and M. Luo, "Calibrated early-warning models with fairness auditing and selective prediction for course withdrawal risk: Evidence from OULAD," *PLOS ONE*, vol. 21, no. 7, p. e0352867, 2026.

[6] D. Gašević, S. Dawson, T. Rogers, and D. Gašević, "Learning analytics should not promote one size fits all: The effects of instructional conditions in predicting academic success," *The Internet and Higher Education*, vol. 28, pp. 68–84, 2016.

[7] R. Conijn, C. Snijders, A. Kleingeld, and U. Matzat, "Predicting student performance from LMS data: A comparison of 17 blended courses using Moodle LMS," *IEEE Transactions on Learning Technologies*, vol. 10, no. 1, pp. 17–29, 2017.

[8] J. López-Zambrano, J. A. Lara, and C. Romero, "Towards portability of models for predicting students' final performance in university courses starting from Moodle logs," *Applied Sciences*, vol. 10, no. 1, p. 354, 2020.

[9] M. Adnan *et al.*, "Predicting at-risk students at different percentages of course length for early intervention using machine learning models," *IEEE Access*, vol. 9, pp. 7519–7539, 2021.

[10] M. Hlosta, Z. Zdrahal, and J. Zendulka, "Ouroboros: Early identification of at-risk students without models based on legacy data," in *Proc. 7th Int. Learning Analytics and Knowledge Conf. (LAK'17)*. ACM, 2017, pp. 6–15.

[11] S. M. Jayaprakash, E. W. Moody, E. J. M. Lauría, J. R. Regan, and J. D. Baron, "Early alert of academically at-risk students: An open source analytics initiative," *Journal of Learning Analytics*, vol. 1, no. 1, pp. 6–47, 2014.

[12] M. Riestra-González, M. d. P. Paule-Ruíz, and F. Ortín, "Massive LMS log data analysis for the early prediction of course-agnostic student performance," *Computers & Education*, vol. 163, p. 104108, 2021.

[13] S. Boyer and K. Veeramachaneni, "Transfer learning for predictive models in massive open online courses," in *Artificial Intelligence in Education (AIED 2015)*, ser. Lecture Notes in Computer Science, vol. 9112. Springer, 2015, pp. 54–63.

[14] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, vol. 45, no. 1, pp. 521–530, 2012.

[15] K. J. M. Janssen, K. G. M. Moons, C. J. Kalkman, D. E. Grobbee, and Y. Vergouwe, "Updating methods improved the performance of a clinical prediction model in new patients," *Journal of Clinical Epidemiology*, vol. 61, no. 1, pp. 76–86, 2008.

[16] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, p. 230, 2019.

[17] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. MIT Press, 1999, pp. 61–74.

[18] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002.

[19] Z. C. Lipton, Y.-X. Wang, and A. J. Smola, "Detecting and correcting for label shift with black box predictors," in *Proc. 35th Int. Conf. on Machine Learning (ICML 2018)*, 2018, arXiv:1802.03916.

[20] Y. Li, N. Wang, J. Shi, X. Hou, and J. Liu, "Adaptive batch normalization for practical domain adaptation," *Pattern Recognition*, vol. 80, pp. 109–117, 2018.

[21] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Int. Conf. on Learning Representations (ICLR 2021)*, 2021, arXiv:2006.10726.

[22] J. Gardner, C. Brooks, and R. Baker, "Evaluating the fairness of predictive student models through slicing analysis," in *Proc. 9th Int. Conf. on Learning Analytics & Knowledge (LAK'19)*. ACM, 2019, pp. 225–234.

[23] R. S. Baker and A. Hawn, "Algorithmic bias in education," *International Journal of Artificial Intelligence in Education*, vol. 32, no. 4, pp. 1052–1092, 2022.

[24] R. A. Opoku, B. Pei, and W. Xing, "Unveiling accuracy-fairness trade-offs: Investigating machine learning models in student performance prediction," *Journal of Learning Analytics*, vol. 12, no. 2, pp. 125–139, 2025.

[25] A. Chouldechova, "Fair prediction with disparate impact: A study of bias in recidivism prediction instruments," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.

[26] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent trade-offs in the fair determination of risk scores," in *8th Innovations in Theoretical Computer Science Conf. (ITCS 2017)*, ser. LIPIcs, vol. 67, 2017, pp. 43:1–43:23.

[27] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, 2016, pp. 3315–3323.

[28] E. H. Simpson, "The interpretation of interaction in contingency tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 13, no. 2, pp. 238–241, 1951.

[29] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[30] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[31] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[32] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.

[33] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774.

[34] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.